DATASET INSPECTION REPORT

Telco Customer Churn Dataset

Data Loading, Quality Controls & Descriptive Statistics

Company	ABC Communications Ltd
Programme	AnalystLab Africa — Data Analytics Internship Programme
Project	Week 1 — Business Analytics Case Study
Dataset	WA_Fn-UseC_-Telco-Customer-Churn.csv

Prepared by Nancy Lee YIMBERE ALAPINI

Performance & Decision Intelligence Analyst | KPI Design, Performance Management & Decision Support | Telecom QoS • Human & Operational Performance

1. Inspection Objective

This report documents the loading and inspection of the Telco Customer Churn dataset before business analysis. In accordance with the AnalystLab Africa Week 1 assignment, the inspection covers the number of rows and columns, data types, missing values, duplicate records and summary statistics.

The objective is to establish whether the dataset is structurally sound and analytically usable, identify data-quality issues that require treatment, and document the preparation decisions applied before churn analysis.

2. Dataset Structure

Inspection item	Result
Rows	7,043
Columns	21
Unit of observation	One customer per row
Unique identifier	customerID
Target variable	Churn
Target categories	Yes / No

The dataset has customer-level granularity. Each row represents one customer and the 21 columns describe customer profile, tenure, subscribed services, contractual and payment characteristics, charges, and churn status.

3. Variable Families and Data Types

The source file contains predominantly categorical variables, together with a small set of numerical fields and one customer identifier. The original imported data types were inspected before transformation.

Variable family	Variables	Inspection note
Identifier	customerID	Text identifier; not an explanatory measure.
Customer profile	gender, SeniorCitizen, Partner, Dependents	SeniorCitizen is stored as integer (0/1) but is conceptually binary categorical.
Customer relationship	tenure	Integer; customer tenure in months.
Subscribed services	PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies	Categorical service-status fields.
Contract & billing	Contract, PaperlessBilling, PaymentMethod	Categorical commercial and payment fields.
Financial	MonthlyCharges, TotalCharges	MonthlyCharges is numeric; TotalCharges is imported as text because of blank entries.
Target	Churn	Categorical target: Yes / No.

4. Missing-Value Inspection

A standard null-value check returns no explicit nulls in the raw CSV. However, a deeper inspection of TotalCharges identifies 11 blank-string values. When TotalCharges is converted to a numeric field, these 11 blanks become missing values.

Field	Missing / blank values	Share of dataset
TotalCharges	11	0.16%
All other fields	0	0.00%

4.1 Business Meaning of the 11 Missing TotalCharges Values

All 11 customers with a blank TotalCharges value have tenure = 0. This pattern is therefore not random: it is consistent with newly acquired customers who have not yet accumulated historical charges. In business terms, the missing TotalCharges values represent customers at the very beginning of the customer lifecycle rather than evidence of a duplicated customer or an obvious service-category error.

For analytical preparation, the blank strings should first be converted to proper missing values and TotalCharges converted to numeric. These 11 records should not be deleted automatically, because doing so would remove valid new customers and could bias early-life churn analysis.

Decision rule: retain the 11 customer records, preserve the missing status of TotalCharges for analyses requiring that field, and exclude/handle those missing values explicitly in numerical calculations.

5. Duplicate Inspection

Control	Result
Fully duplicated rows	0
Duplicated customerID values	0
Distinct customerID values	7,043

The uniqueness control confirms that each customer is represented once. This protects customer counts, churn rates and segment-level calculations from duplication bias.

6. Summary Statistics

Descriptive statistics were calculated for the principal continuous numerical variables. TotalCharges statistics are based on 7,032 valid numeric observations after the 11 blank values are treated as missing.

Variable	Count	Mean	Std. Dev.	Min	25%	Median	75%	Max
tenure	7,043	32.37	24.56	0	9	29	55	72
MonthlyCharges	7,043	64.76	30.09	18.25	35.50	70.35	89.85	118.75
TotalCharges	7,032	2,283.30	2,266.77	18.80	401.45	1,397.48	3,794.74	8,684.80

6.1 Statistical Reading

- tenure ranges from 0 to 72 months, confirming that the dataset includes both newly acquired and long-standing customers.
- MonthlyCharges range from 18.25 to 118.75, indicating substantial variation in monthly billing levels across customer configurations.
- TotalCharges range from 18.80 to 8,684.80 among valid observations. Because this variable is cumulative, its interpretation depends on both monthly charges and customer tenure.
- SeniorCitizen is stored numerically but should not be interpreted using continuous-variable statistics because it is a binary indicator.

7. Business Consistency Controls

Technical inspection was complemented by business-rule checks to avoid treating valid business categories as data-quality problems.

Tenure and TotalCharges. The 11 blank TotalCharges records all correspond to tenure = 0, which supports the interpretation that these are new customers without accumulated historical charges.

Service-status categories. Values such as 'No internet service' and 'No phone service' are valid business categories. They must not be recoded as missing values.

Churn target. The target field contains the expected Yes / No business categories.

Customer identifier. customerID is unique across all 7,043 rows and can therefore be used as the customer-level counting key.

8. Data Preparation Decisions

- Convert blank TotalCharges strings to missing values and convert TotalCharges to a numeric data type.
- Retain the 11 tenure = 0 customer records rather than deleting them automatically.
- Handle the 11 missing TotalCharges values explicitly in analyses that require this variable.
- Preserve valid service-status categories such as 'No internet service' and 'No phone service'.
- Use customerID as the unique customer key and verify its uniqueness before customer-count calculations.
- Create analytical derived fields, such as tenure groups, separately from the original source fields so that source-data integrity is preserved.

9. Dataset Quality Assessment

Quality dimension	Assessment	Evidence
Structure	Satisfactory	7,043 customer-level rows and 21 fields.
Uniqueness	Satisfactory	0 duplicated rows; 0 duplicated customerID values.
Completeness	Satisfactory with one controlled issue	11 blank TotalCharges values (0.16%), all linked to tenure = 0.
Data types	Minor transformation required	TotalCharges must be converted from text to numeric.
Categorical consistency	Satisfactory	Service-status categories are meaningful business values.
Analytical usability	Good after preparation	Dataset is suitable for churn segmentation

and statistical analysis after the documented TotalCharges treatment.

10. Inspection Conclusion

The Telco Customer Churn dataset is structurally suitable for the Week 1 business analytics case study. The inspection confirms 7,043 rows, 21 columns, no duplicated rows and no duplicated customer identifiers.

The principal data-quality issue is limited to 11 blank TotalCharges values, representing 0.16% of the dataset. All 11 records have tenure = 0, providing a coherent business explanation: these are customers at the start of their relationship who have not yet accumulated historical charges. The appropriate response is therefore controlled treatment rather than automatic record deletion.

After converting TotalCharges to numeric and handling these 11 missing values explicitly, the dataset is sufficiently complete, consistent and analytically usable for churn segmentation, visualisation and business insight generation.